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Physical, chemical and isotopic characteristics of groundwater and surface water in the Lake Chilwa Basin, Malawi

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ABSTRACT

We measured the physical, chemical and isotopic properties of groundwater and surface water from streams and Lake Chilwa in the Lake Chilwa Basin in southern Malawi. The study objective was to assess water quality and determine the cause of water quality deterioration. Groundwater from 16 boreholes, 5 stream samples and 3 samples from Lake Chilwa were investigated for temperature, pH, electrical conductivity, total dissolved solids (TDS), major ions and stable isotopes of hydrogen and oxygen. Groundwater and water from Lake Chilwa had higher ionic concentrations (e.g., Cl^-, HCO_3^-, Na^+, Ca^{2+}, Mg^{2+}) than stream water. Ionic evolution models indicate weathering of silicate minerals as the major control of the ionic concentrations in groundwater. The chemistry of water in Lake Chilwa is markedly affected by evaporation. The concentrations of the TDS, HCO_3^-, F^- and NO_3^- in groundwater at several locations were above the recommended limits prescribed for drinking water by the World Health Organization. The results of this study show that poor groundwater quality for drinking and domestic use is due to natural water-rock interactions (TDS, HCO_3^-, F^-) and anthropogenic pollution (NO_3^-). Because stream water is of good quality, and streamflow is supported by baseflow from fresh groundwater, efforts should be made to map fresh groundwater aquifers for domestic, agricultural and industrial use.

1. Introduction

Groundwater and surface water are the major sources of water supply for domestic, agricultural and industrial use. Approximately one third of the world's population depend on groundwater for domestic use (Nickson et al., 2005). The World Health Organization (WHO) reports that having safe drinking water is an internationally accepted human right (WHO, 2004). However, groundwater resources are under constant threat of depletion and salinization, which is a worldwide concern (e.g., Rajmohan and Elango, 2004; Gibrilla et al., 2010).

A report by the British Geological Survey (BGS) (BGS, 2004) indicates that groundwater in some parts of Malawi is characterized by high salinity as a result of evaporative concentration or dissolution of evaporite minerals in sedimentary rocks. The occurrence of some groundwater with high salinity in southern Malawi is a major water quality concern (e.g., Monjerezi and Ngongondo, 2012; Monjerezi et al., 2011). Boreholes which tap groundwater with high salinity are frequently abandoned in favor of open unprotected shallow hand dug holes in dry river beds or surface water (Bath, 1980). However, the expansion of groundwater supply countrywide and the improvement of drinking water quality remains a national priority of the government of Malawi (GOM) (Mkandawire, 2004; GOM, 2007).

Previous investigations of the quality of groundwater and surface water in the Lake Chilwa Basin in southern Malawi focused on determining the physical properties and concentrations of chemical species and biological pollutants and the conformance of the water to WHO and Malawi Bureau of Standards (MBS) drinking water quality guidelines (Saka and Ambali, 1999; Palamuleni, 2002; Basin and Saka, 2006; Chavula and Mulwafu, 2007; Msonda et al., 2007; Pritchard et al., 2008; Von Hellens, 2013). Although groundwater from boreholes are used by a majority of the population in Malawi, information on the chemical composition of the water in these boreholes is either scanty or non-existent (Bath, 1980; Chilton and Smith-Carington, 1984; McFarlane and Bowden, 1992; Palamuleni, 2002; Sajidu et al., 2008; Kanyerere et al., 2010; Chidya et al., 2016). There are no dedicated studies focused on the hydro-geochemical properties designed to understand the processes that control the quality of groundwater and surface water in the Lake Chilwa Basin.

Geochemical processes occurring within the groundwater and surface water, and reactions with aquifer minerals have a profound effect

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on water quality. These geochemical processes are responsible for the temporal and spatial variations in groundwater and surface water chemistry and quality. The stable isotopes of oxygen ($\delta^{18}O$) and hydrogen (δD) of water allow rain that falls on land to be traced through surface and groundwater reservoirs. The partitioning of the isotopes of hydrogen and oxygen during physical processes such as evaporation or chemical processes that involve water-rock interaction makes the $\delta^{18}O$ and δD useful in the understanding of hydro-chemical evolution of groundwater and surface water (e.g., Gat and Tzur, 1968; Ganyaglo et al., 2017). The fact that the previous studies in the Lake Chilwa Basin did not use the $\delta^{18}O$ and δD of water to trace the chemical evolution of groundwater and surface water in relation to water quality is a major knowledge gap in our understanding of the factors that control water quality in the Lake Chilwa Basin. The objective of this study was to assess water quality and determine the cause of water quality deterioration. We used a combination of the physical properties, the major ion concentrations and $\delta^{18}O$ and δD of water and their relation to water-rock interaction to determine the dominant geochemical processes responsible for groundwater and surface water quality. The results of this study bridge the information gap needed to address groundwater and surface water quality issues, as well as provide a foundation for groundwater quality monitoring.

2. Study area

2.1. Location and physiography

Lake Chilwa Basin (latitude 15° 20'S, longitude 35° 40'E) is a closed basin of about 7500 km^2 located in southern Malawi and includes portions of northeastern Mozambique (Fig. 1). The Lake Chilwa Basin covers three administrative districts: Phalombe has about 73,880 ha, Machinga has about 11,200 ha and Zomba has about 185,174 ha of land (Sagona et al., 2016). The Lake Chilwa Basin is bounded to the west by the Chikala Hills, Zomba Mountain, Malosa Mountain, Shire Highlands and Chiradzulu Mountain. These mountains and highlands give rise to the Domasi River, Likangala River, Thondwe River and the Namadzi River. The Lake Chilwa Basin is bounded to the southeast by the Mulanje Mountain and the Michesi Mountain which give rise to the Phalombe River and Sombani River. To the east lie the hills and mountains of Mozambique which give rise to Mnembo River and Mbugwe River (Njaya, 2001a,b).

2.2. Geology

The geology of the Lake Chilwa Basin consists of a Basement Complex of crystalline metamorphic and igneous rocks of Precambrian to lower Paleozoic age (Garson, 1960; Carter and Bennett, 1973; Chilton and Smith-Carington, 1984; Mapoma and Xie, 2014). The major lithological units of the basement complex are Charnockites and granulites (Bloomfield, 1965). The Charnockites include orthopyroxene-bearing quartz and feldspar, whereas the granulites are rich in pyroxene, biotite, garnet, calcium plagioclase and quartz. The Basement Complex is intruded by alkaline igneous rocks consisting of basanite, granite, nepheline syenite and carbonatite. The nepheline syenite consists largely of nepheline and alkali feldspars which include microcline, albite, mica, hornblende and magnetite. The carbonatites consists of calciocarbonate, magnesiocarbonate and ferrocarbonate. The study by Swanzie and Stubbs (1972) show that the carbonatites contain rare earth elements, apatite and carbonates. Cenozoic alluvial fans and lacustrine sediments cover the valleys and the lowlands (Bloomfield, 1965).

The Lake Chilwa Basin is a tectonic depression of post-cretaceous age filled with sand, silt and various sediments from the denudation of the surrounding highlands (Sagona et al., 2016). The uplands have deep, well drained sandy soils derived from the weathering of crystalline rocks, while the lowlands have thick soils developed in fluvial, colluvial and lacustrine deposits (Morgan and Kalk, 1970).

2.3. Climate

The Lake Chilwa Basin experiences a warm tropical climate, which is characterized by a rainy and dry season. The rainy and dry seasons occur from November to April and May to October, respectively. The mean annual temperature is 24 °C. The highest temperature of 29 °C and the lowest temperature of 20 °C are experienced in October and June, respectively.

2.4. Hydrogeology

The Lake Chilwa Basin (Fig. 1) has a radial drainage pattern (Lancaster, 1979). Lake Chilwa is at an altitude of 650 m above mean sea level (amsl). The maximum depth of the lake is 2.7 m and the lake surface measures 1759 km^2. There are six major influent rivers draining into Lake Chilwa: Domasi, Likangala, Thondwe, Phalombe, Sombani and Mnembo with maximum discharges which occur during the rainy season from December to April, usually peaking in February (Lancaster, 1979). The Mnembo River flows from Mozambique into Lake Chilwa and is actually the lake's largest source of recharge (Jamu et al., 2003). Rainfall variability impacts water availability and the discharge of streams in particular catchments (Bloomfield, 1965; McCarthy, 2001; Mapoma and Xie, 2014).

The aquifers in the Lake Chilwa Basin include surficial aquifers consisting of alluvial and extensively weathered basement rocks and deep fractured crystalline basement aquifers (Pritchard et al., 2008; Mapoma and Xie, 2014). The surficial aquifers are 15–30 m thick (Chilton and Smith-Carington, 1984) and consist of residual soils (latosols) at the surface, an upper saprolite layer, a lower saprolite layer and a saprock that overlies fresh bedrock (Acworth, 1987; Wright, 1992). The mineralogy of the surficial aquifers includes iron oxide, laterites, calcretes and clays (Bloomfield, 1965; Acworth, 1987; Wright, 1992). The deep aquifers of the Basement Complex are spatially variable in thickness and consists of variably fractured crystalline metamorphic and igneous rocks (e.g., Bloomfield, 1965; Chilton and Smith-Carington, 1984; Acworth, 1987; Wright, 1992).

The rivers, lakes, dams, lagoons and marshes connect with groundwater in recharge zones or in discharge zones (GOM, 2011). The crystalline basement aquifers are low yielding because of the less porous nature of fractured igneous and metamorphic rocks, whereas, the shallow surficial aquifers are potentially higher yielding (e.g. Chilton and Smith-Carington, 1984; GOM, 2007; Pritchard et al., 2008). The yields of the surficial aquifers can exceed of 10 L/s (GOM, 2007; Chimphamba et al., 2009). Borehole yields at the higher elevations are highest where the saturated thickness of the weathered zone is greatest (Chilton and Smith-Carington, 1984). The water table which is closer to the surface in the valley region and deeper at higher elevations (Mkandawire, 2008). McFarlane and Bowden (1992) indicates that the shallow aquifers below very flat land at least 1 km from the hills or mountains are likely to be higher yielding than shallow aquifers closer to the hills.

3. Methods

Water samples for this study were collected from boreholes, streams and Lake Chilwa using standard sampling procedures (APHA, 1985). Water sampling was conducted from 20th December, 2016 to 2nd January, 2017. Groundwater samples were collected from the following 16 boreholes: Dzinja, Namthiya, Nthumbulu, Nambwale, Namuli, Chambeni, Nanthenga, Mandawala, Migowi, Likapa, Govala, Kachoka, Pirimiti, Jali police, Lomoni and Kumpalila (Fig. 1). All the boreholes were drilled into the fractured basement, which we consider the “deep” aquifer, compared to the “shallow” aquifers composed of regolith and alluvium, which overlie the fractured basement aquifer. Groundwater

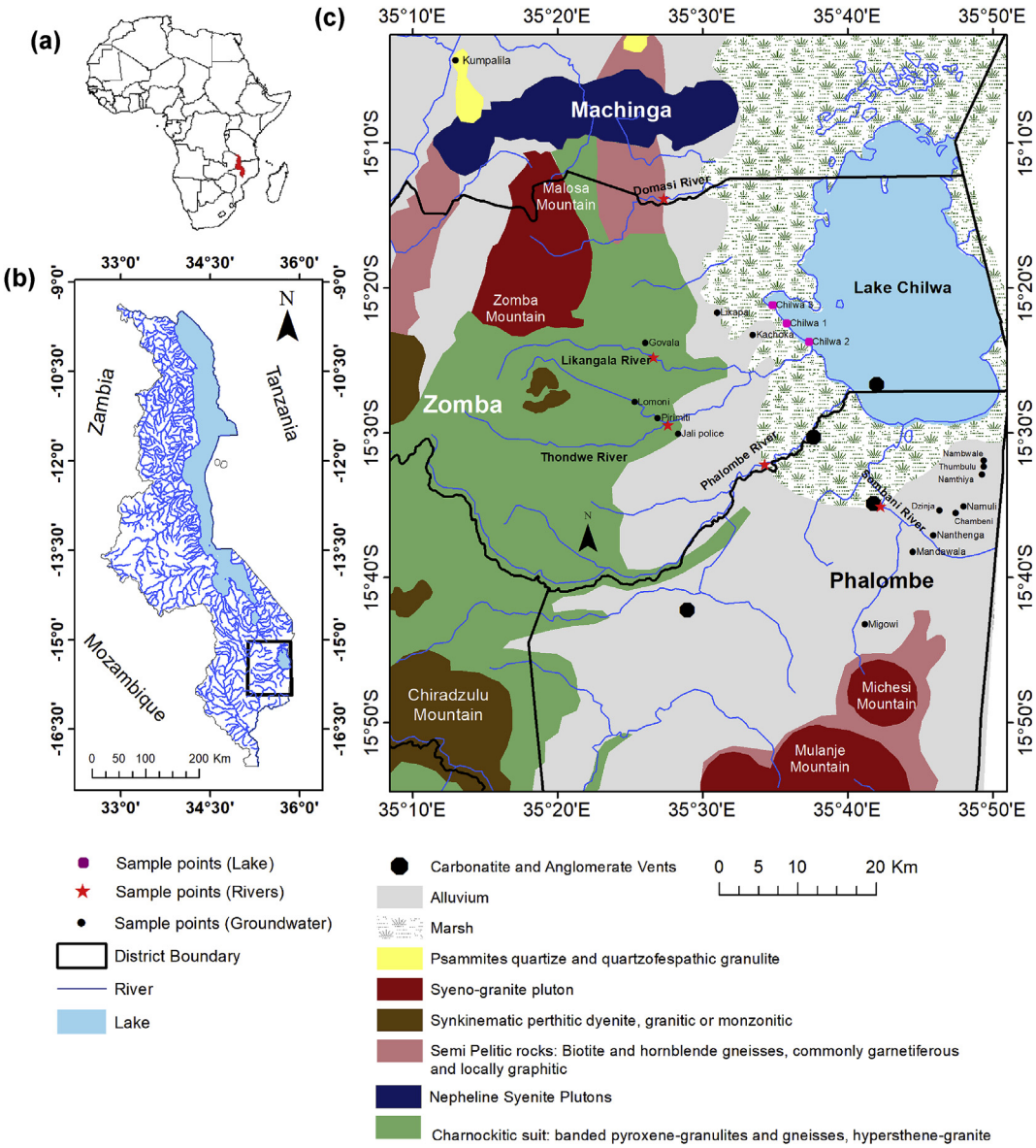

Fig. 1. Map of (a) Africa showing the location of Malawi, (b) Malawi showing drainage basins with the rectangle showing the location of the study area and (c) the study are in Lake Chilwa Basin showing the geology and groundwater and surface water sampling locations. Shape files obtained from the Geological Survey Department, Malawi.

samples were collected by hand pumping after purging an estimated three casing volumes.

All stream samples were collected at easily accessed points (e.g., under bridges) from the following streams: Domasi, Likangala, Thondwe, Phalombe and Sombani (Fig. 1). Lake samples were collected from the western side of Lake Chilwa along the shore from three locations (Fig. 1). Stream and lake samples were collected by the grab technique.

All water samples were filtered through 0.45 μM nylon filters during collection. The sample bottles were rinsed three times with the filtered sample to be collected and then filled and capped. Filtered water for cations analyses were collected in 60 mL polypropylene bottles, acidified to a pH < 2 with trace metal grade nitric acid (APHA, 1985) and stored in a cooler box. Filtered samples for anion analysis were collected unacidified in 30 mL polypropylene bottles and stored in a cooler box. Water samples for dissolved inorganic carbon (DIC) were collected using syringes and introduced into vacutainer tubes pre-loaded with 1.5 mL of 85% phosphoric acid and magnetic stir bars (Atekwana and Krishnamurthy, 1998). Water for δD and $\delta^{18}O$ was collected and stored in 25 mL scintillation vials with inverted cone closures to prevent evaporation. The samples were store in the laboratory until analyses.

Water temperature, pH, electrical conductivity (EC) and total dissolved solids (TDS) were measured insitu at each borehole during purging using potable multi-meter probe (HANNA instrument model 991300N). The probe of the meter was dipped in the container receiving the pumped water and readings were taken after stabilization (APHA, 1985). For lake and stream water, the probe was immersed in the water and readings were recorded after stabilization.

In the field, alkalinity was determined by H_2SO_4 titration to an acid equivalent endpoint of 4.2 using a digital titrator (Hach Company, 1992). Major anions and cations (F^-, Cl^-, SO_4^{2-}, NO_3^- and Na^+, K^+, Ca^{2+}, Mg^{2+}) were determined by ion chromatography. Dissolved inorganic carbon concentrations were calculated from carbon dioxide gas extracted and cryogenically purified in a vacuum line (Atekwana and Krishnamurthy, 1998). Stable isotopes of hydrogen and oxygen of the water samples were measured using an Off Axis-Integrated Cavity Output Spectroscopy manufactured by Los Gatos and sold as the Triple Liquid Water Analyzer. The samples were analyzed at Western

Table 1
Summary statistics of the physical and chemical compositions of water samples. All values are in mg/L except pH, temperature (°C), electrical conductivity (EC) (μS/cm) and oxygen isotope ($\delta^{18}O$) (‰) and hydrogen isotope (δD) (‰). GW is groundwater, LW is lake water, SW is stream water, Min is minimum value, Max is maximum value and SD is the standard deviation.

		pH	Temp	EC	TDS	Na^+	K^+	Mg^{2+}	Ca^{2+}	F^-	Cl^-	NO_3^-	SO_4^{2-}	HCO_3^-	$\delta^{18}O$	δD
GW	Min	6.7	27.0	1563	1000	52.2	0.2	13.9	36.4	0.8	21.3	27.3	0.2	286	−6.3	−39
	Max	7.2	30.1	3497	2238	149.9	3.4	42.1	122.4	39.5	134.7	59.3	21.9	488	−4.7	−28
	Mean	6.9	28.1	2450	1568	92.8	1.5	28.9	79.8	11.1	78.5	41.4	6.2	348	−5.8	−35
	SD	0.2	0.9	604	387	33.3	0.9	7.1	25.2	11.8	38.1	12.3	6.1	50	0.5	3
LW	Min	8.4	27.3	3001	1921	319.0	12.0	6.0	17.4	40.4	320.3	14.1	17.0	253	5.5	33
	Max	8.8	27.8	3589	2297	338.0	12.7	10.7	20.6	42.0	363.2	18.8	19.7	297	8.0	45
	Mean	8.6	27.5	3313	2121	328.4	12.4	7.9	19.0	41.4	343.0	16.3	18.2	276	7.0	40
	SD	0.2	0.3	296	189	9.5	0.4	2.5	1.6	0.9	21.6	2.4	1.4	22	1.3	6
SW	Min	7.3	22.3	864	553	10.6	2.5	1.3	1.8	0.6	2.4	4.0	0.5	26	−5.0	−30
	Max	7.9	29.5	953	610	42.2	4.4	2.3	3.3	2.6	8.1	12.6	10.4	111	2.8	17
	Mean	7.5	26.4	924	592	23.3	3.4	1.9	2.5	2.0	5.4	10.4	2.9	63	−1.9	−10
	SD	0.2	3.2	37	23	11.6	0.8	0.5	0.6	0.8	2.3	3.6	4.2	35	3.4	20

Michigan University, Kalamazoo Michigan USA. Stable isotope ratios for hydrogen and oxygen are reported in the delta (δ) notation in per mil (‰):

$$\delta\ (‰) = ((R_{sample}/R_{standard}) - 1) \times 1000$$

Where R is $^2H/^1H$ or $^{18}O/^{16}O$. The δ values for H and O isotopes are reported relative to the Vienna Standard Mean Ocean Water (VSMOW). Routine isotopic measurements of in-house standards and samples have an overall precision of 0.2‰ and 1‰ for O and H isotopes, respectively.

The results of the physical-chemical analyses were uploaded to AquaChem (Waterloo Hydrologic) where water types were determined. Phreeqc version 2 was used to calculate mineral saturation indices using pH, DIC, Ca^{2+} and temperature (Parkhurst and Appelo, 1999).

4. Results

Summary statistics of the results are presented in Table 1, while the results of the physical, chemical and isotopic parameters are presented in Table 2.

4.1. Physical parameters

The pH of groundwater ranged from 6.8 to 7.2 and averaged 6.9 ± 0.2. The pH of lake samples were more alkaline and ranged from 8.4 to 8.8 and averaged 8.6 ± 0.2. Stream samples had an average pH of 7.5 ± 0.2 and ranged from 7.4 to 7.9. The temperature of groundwater, lake and stream samples ranged from 27.1 to 30.1 °C, 27.3–27.8 °C and 22.3–29.5 °C, respectively and averaged 28.1 ± 0.9, 27.5 ± 0.3 and 23.3 ± 3.2 °C, respectively. The EC of groundwater, lake and stream samples ranged from 1563 to 3497, 3001 to 3589 and 1350 to 1489 μS/cm, respectively and averaged 2450 ± 604, 3313 ± 296 and 1445 ± 58 μS/cm, respectively. The TDS of groundwater, lake and stream samples ranged from 1000 to 2238, 1921 to 2297 and 553–610 mg/L, respectively and averaged 1568 ± 387, 2121 ± 189 and 592 ± 23 mg/L, respectively.

4.2. Chemical parameters

The concentrations of Na^+, K^+, Mg^{2+} and Ca^{2+} for groundwater samples ranged from 52.2 to 149.9, 0.4 to 3.4, 13.9 to 42.1 and 36.4–122.4 mg/L and averaged 92.8 ± 33.3, 1.5 ± 0.9, 28.9 ± 7.1 and 79.8 ± 25.2 mg/L, respectively. For lake samples, the concentrations of Na^+, K^+, Mg^{2+} and Ca^{2+} ranged from 319.0 to 338.0, 12.0 to 12.7, 6.0 to 10.7 and 17.4–20.6 mg/L and averaged 328.4 ± 9.5, 12.4 ± 0.4, 7.9 ± 2.3 and 19.0 ± 2.0 mg/L, respectively. For stream samples, the concentrations of Na^+, K^+, Mg^{2+} and Ca^{2+} ranged from 10.6 to 42.2, 3.0 to 12.4, 1.3 to 2.3 and 1.8–3.5 mg/L and averaged 23.3 ± 11.6, 4.6 ± 3.3, 1.7 ± 0.5 and 2.5 ± 0.6 mg/L, respectively. The order of abundance of the cations for the groundwater is $Na^+ > Ca^{2+} > Mg^{2+} > K^+$, for lake water is $Na^+ > Ca^{2+} > K^+ > Mg^{2+}$ and for stream water is $Na^+ > K^+ > Ca^{2+} > Mg^{2+}$.

The concentrations of F^-, Cl^-, NO_3^-, SO_4^{2-} and HCO_3^- in groundwater samples ranged from 0.8 to 39.5, 21.3 to 134.7, 27.3 to 59.3, 0.2 to 21.9 and 286.0–428.0 mg/L, respectively and averaged 11.1 ± 11.8, 78.5 ± 38.1, 41.4 ± 12.3, 6.2 ± 6.1 and 344.7 ± 40.7 mg/L, respectively. For lake samples, the concentrations of F^-, Cl^-, NO_3^-, SO_4^{2-} and HCO_3^- ranged from 40.4 to 42.0, 320 to 363, 14.1 to 18.8, 17.0 to 19.7 and 253.0–297.1 mg/L, respectively and averaged 41.4 ± 0.9, 343 ± 22, 16.3 ± 2.4, 18.2 ± 1.4 and 275.5 ± 22.1 mg/L, respectively. For stream samples, the concentrations of F^-, Cl^-, NO_3^-, SO_4^{2-} and HCO_3^- ranged from 0.6 to 2.6, 2.4 to 8.1, 4.0 to 12.6, 0.5 to 10.4 and 26.2–111.0 mg/L, respectively and averaged 2.0 ± 0.8, 5.4 ± 2.3, 10.4 ± 3.6, 2.9 ± 4.2 and 63.1 ± 34.9 mg/L, respectively. The order of anion abundance is $HCO_3^- > Cl^- > NO_3^- > F^- > SO_4^{2-}$ for groundwater, $Cl^- > HCO_3^- > F^- > SO_4^{2-} > NO_3^-$ for lake water and $HCO_3^- > NO_3^- > Cl^- > SO_4^{2-} > F^-$ for stream water.

4.3. Isotopic parameters

The $\delta^{18}O$ composition of groundwater, lake and stream samples ranged from −6.3 to −4.7, 5.5 to 8.0 and −5.0 to 2.8‰, respectively and averaged −5.8 ± 0.5, 7.0 ± 1.3 and −1.9 ± 3.4‰, respectively. The δD composition of groundwater, lake and stream samples range from −39 to −28, 33 to 45 and -30 to 17‰, respectively, and averaged of −35 ± 3, 40 ± 6 and −10 ± 20‰, respectively.

5. Discussion

5.1. Water quality of groundwater and surface water

The quality of groundwater and surface water determines its suitability for different purposes depending upon specific standards. The maximum permissible level (MPL) for chemical constituents in drinking water (Table 3) by WHO (2008) and the MBS (2005) are the basis for the water quality evaluation.

5.1.1. Physical properties

The pH values of groundwater in the Lake Chilwa Basin which range from 6.8 to 7.5 is slightly acidic to slightly alkaline and the pH of lake water and stream water are alkaline, with ranges of 8.4–8.8 and 7.4 to

Table 2
Results of the physical, chemical and isotopic analyses of water samples from Lake Chilwa Basin (- = not measured).

SampleID	StationID	pH	Temp. (°C)	EC (μS/cm	TDS (mg/L)	Na^+ (mg/L)	K^+ (mg/L)	Mg^{2+} (mg/L)	Ca^{2+} (mg/L)	F^- (mg/L)	Cl^- (mg/L)	NO_3^- (mg/L)	SO_4^{2-} (mg/L)	HCO_3^- (mg/L)	DIC (mg C/L)	$\delta^{18}O$ (‰)	δ^2H (‰)	SI calcite	SI dolomite	SI gypsum
Pt 1 BH	Dzinja	6.88	27.1	2861	1831	149.9	0.7	28.9	122.4	25.4	134.7	37.7	2.6	417.9	115	−6.3	−38	−0.4	−0.7	−2.9
Pt 2 BH	Namthiya	6.84	27.3	2780	1779	126.2	1.7	34.1	119.2	5.56	130.3	35.4	9.5	488.0	189	−6.3	−37	−0.2	−0.6	−2.7
Pt 3 BH	Thumbulu	6.93	30.1	2869	1836	138.2	2.7	19.3	88.8	15.2	132.6	57.4	10.3	381.9	121	−6.1	−35	−0.1	−0.6	−2.6
Pt 4 BH	Nambwale	6.78	28.9	1948	1247	115.7	0.8	36.7	122.2	11.4	129.3	59.3	13.5	341.6	109	−6.0	−35	−0.1	−0.6	−2.7
Pt 5 BH	Namuli	6.71	28.2	3497	2238	95.0	2.1	42.1	80.0	32.1	104.5	41.6	6.8	390.3	148	−6.0	−36	0.0	−0.2	−2.8
Pt 6 BH	Chambeni	7.10	28.9	3206	2052	128.7	1.8	25.8	76.4	17.1	96.8	56.7	9.5	357.1	199	−6.1	−38	−0.2	−0.5	−2.7
Pt 7 BH	Nanthenga	6.85	28.5	2815	1802	130.7	1.7	32.4	100.2	39.5	76.0	28.9	1.3	313.9	210	−5.9	−36	−0.2	−0.6	−3.8
Pt 8 BH	Mandawala	7.11	29.8	3100	1984	61.7	1.9	33.9	66.2	5.8	62.6	59.0	4.7	326.4	208	_	_	−0.2	−0.4	−3.0
Pt 9 BH	Migowi	6.82	27.9	1755	1123	82.2	1.8	13.9	68.4	1.1	47.8	58.6	1.1	322.4	146	_	_	−0.4	−1.0	−3.7
Pt 10 BH	Likapa	7.06	27.6	1563	1000	75.7	0.4	28.0	80.4	3.1	77.2	31.4	0.3	286.0	131	_	_	−0.5	−0.8	−4.3
Pt 11 BH	Govala	6.79	27.0	1610	1030	66.9	2.0	20.2	67.5	0.8	61.4	34.6	0.8	316.8	135	−5.6	−33	−0.5	−1.1	−3.9
Pt 12 BH	Kachoka	6.88	28.0	2359	1510	62.6	0.2	26.7	52.4	2.8	33.2	37.2	1.6	340.6	133	−4.7	−28	−0.4	−0.7	−3.7
Pt 13 BH	Pirimiti	6.95	27.3	1962	1256	52.2	0.9	33.2	68.4	1.8	44.8	37.4	0.2	300.6	129	−5.0	−31	−0.4	−0.7	−4.4
Pt 14 BH	Jali police	7.11	27.8	2680	1715	77.1	0.4	25.3	36.4	5.7	54.5	29.9	11.3	314.7	140	−5.9	−39	−0.3	−0.6	−2.7
Pt 16 BH	Lomoni	6.70	27.4	2200	1408	52.7	3.4	33.9	71.6	5.6	49.6	27.3	3.0	336.0	127	_	_	−0.4	−0.6	−3.3
Pt 17 BH	Kumpalila	7.22	27.5	1990	1274	69.9	1.6	27.7	55.6	4.7	21.3	29.3	21.9	341.6	142	_	_	−0.6	−0.8	−2.7
Pt 1 SW	L. Chilwa	8.41	27.8	3589	2297	328.1	12.7	10.7	20.6	40.4	363.2	18.8	19.7	276.5	98	8.0	45	−1.3	−2.0	−5.9
Pt 7 SW	L. Chilwa 2	8.78	27.5	3350	2144	338.0	12.4	7.1	19.0	42.0	345.5	16.0	18.0	297.1	106	7.4	43	−1.0	−2.2	−3.4
Pt 8 SW	L. Chilwa 3	8.63	27.3	3001	1921	319.0	12.0	6.0	17.4	41.8	320.3	14.1	17.0	253.0	94	5.5	33	−1.4	−2.5	−2.6
Pt 2 SW	Phalombe R	7.52	28.8	915	586	42.2	3.0	1.3	1.8	2.6	8.1	4.0	10.4	111.0	85	0.5	4	−2.2	−4.3	−4.0
Pt 3 SW	Sombani R	7.42	23.8	949	607	22.9	4.4	2.3	3.3	0.6	6.1	12.6	1.1	86.6	86	2.8	17	−2.6	−5.2	−5.5
Pt 4 SW	Thondwe R	7.93	29.5	941	602	20.5	3.9	2.3	2.3	2.6	6.6	11.7	1.3	39.7	78	−2.8	−14	−2.5	−4.9	−4.7
Pt 5 SW	Likangala R	7.35	27.4	864	553	20.2	3.1	2.1	2.5	2.2	3.8	11.3	1.4	51.9	88	−5	−30	−1.9	−4.1	−4.2
Pt 6 SW	Domasi R	7.32	22.3	953	610	10.6	2.5	1.3	2.8	2.1	2.4	12.4	0.5	26.2	80	−4.8	−25	−1.8	−4.9	−4.4

Table 3
Water quality standards. World health organization (WHO) Drinking water guidelines (WHO, 2008) and Malawi Bureau of standards (MBS) maximum permissible levels (MBS, 2005).

Constituent	Unit	WHO Guideline	MBS Guideline
Total dissolved solid (TDS)	mg/L	1000	1000
Sodium	mg/L	Not stated	200
Potassium	mg/L	10	10
Magnesium	mg/L	150	150
Calcium	mg/L	200	200
Fluoride	mg/L	1.5	2
Chloride	mg/L	250	250
Nitrate	mg/L	50	50
Sulfate	mg/L	400	400
Bicarbonate	mg/L	300	Not stated

7.9, respectively. The pH values of the water in the Lake Chilwa Basin are within the MPL of 6–9 prescribed for drinking water by WHO (2008) and MBS (2005).

High TDS concentrations decrease the palatability of water and causes gastrointestinal irritation to consumers (Garg et al., 2009). Some groundwater samples and lake samples are brackish (TDS 1000–2200 mg/L), whereas, stream water are fresh with TDS < 1000 mg/L (Freeze and Cherry, 1979). Based on the WHO (2008) and MBS (2005) MPL for TDS concentrations in drinking water, 88% of the groundwater samples and all of the lake samples exceed the MPL of 1000 mg/L (Fig. 2a). All stream samples are below the MPL for TDS.

5.1.2. Major cations

Excess Na^+ in water makes it unsuitable for drinking because of severe health problems like hypertension (Holden, 1970). According to the WHO (2008) and MBS (2005) guidelines, the MPL for Na^+ in drinking water is 200 mg/L. Sodium concentrations in all the groundwater and stream samples are below the MPL, while the concentrations of Na^+ in the lake samples exceed the MPL (Fig. 2b), consistent with earlier findings of Basin and Saka (2006). The concentration of K^+ in groundwater samples and stream samples is less than 10 mg/L which is the MPL as per WHO (2008) and MBS (2005). However, the concentration of K^+ in lake samples exceed the MPL of 10 mg/L (Fig. 2c). The concentration of Mg^{2+} in groundwater, lake water and stream water ranges from 1.3 to 42.1 mg/L and are below the WHO (2008) and MBS (2005) MPL of 150 mg/L. All groundwater samples, lake samples and stream samples have Ca^{2+} concentrations that are below the WHO (2008) and MBS (2005) MPL of 200 mg/L.

5.1.3. Major and minor anions

Fluoride is an essential element for humans (WHO, 2004). Deficiency of F^- in drinking water (< 0.5 mg/L) leads to dental caries (Edmunds and Smedley, 1996). High intake of F^- (> 1.5 mg/L) results in physiological disorders, skeletal and dental fluorosis (Latha et al., 1999). Fourteen groundwater samples (88%), all lake samples and four (80%) stream samples (Fig. 3a) have F^- concentrations that exceed the MPL of 1.5 mg/L and 2 mg/L prescribed by WHO (2008) and MBS (2005), respectively. Chloride deteriorates the quality of drinking water at high concentrations (McCarthy, 2004). High concentrations of Cl^- in drinking water causes a salty taste and has a laxative effect to people not accustomed to drinking water with high Cl^- concentrations (Bhardwaj and Singh, 2011). All the lake samples exceeded the MPL for Cl^- in drinking water (Fig. 3b) specified at 250 mg/L by WHO (2008) and MBS (2005). All groundwater samples and stream samples have concentrations that are below the MPL for Cl^-. High NO_3^- intake causes methemoglobinemia commonly known as "blue baby syndrome" in infants (Comly, 1945). Methemoglobinemia occurs when nitrite (NO_2^-), a reduced form of nitrate, interacts with red blood cells and impairs their ability to carry oxygen (Mirvish, 1991). This impairment results in anoxia (deficiency of oxygen in the blood) and cyanosis (blue blood). Only five (31%) groundwater samples exceeded the WHO (2008) and MBS (2005) MPL of 50 mg/L for NO_3^- in drinking water (Fig. 3c). All stream samples and lake samples have NO_3^- concentrations below MPL. Higher concentration of SO_4^{2-} in drinking water is associated with respiratory problems (Subba-Rao, 1998). The concentration of SO_4^{2-} in all the groundwater, lake and stream samples are below the WHO (2008) and by MBS (2005) MPL of 400 mg/L.

Bicarbonate (HCO_3^-) is the dominant anion in groundwater, stream water and lake water. Ninety four percent of groundwater samples have HCO_3^- high concentration that exceed the WHO (2008) MPL of 300 mg/L in the drinking water (Fig. 3d). All the stream samples and lake samples have HCO_3^- concentrations below the MPL.

5.2. Processes responsible for groundwater and surface water quality

5.2.1. Rock and mineral weathering and water-rock interaction

Poor quality of surface water and groundwater in the Lake Chilwa Basin is characterized by high concentrations of TDS, Na^+, K^+, HCO_3^-, F^-, Cl^- and NO_3^- which were above the WHO (2008) and MBS (2005) MPL. In natural waters, high concentrations TDS, Na^+, K^+, HCO_3^-, and F^- are attributed to water-rock interaction, while high concentrations of Cl^- and NO_3^- are rarely attributed to water-rock interaction, as these anions are not major constituents of silicate rocks or not produced during weathering of silicate rocks (e.g., Hem, 1985).

The sources and chemical evolution that control the concentration of the major ions (Na^+, K^+, Ca^{2+}, Mg^{2+}, HCO_3^-, SO_4^{2-} and Cl^-) in natural waters can be determined from ionic evolutionary model. Variable sources and processes that control the proportion of the major ions can be determined using a Piper diagram (Fig. 4). We distinguish three dominant water types. The water types have facies of: (A) Ca–Mg–Na + K–HCO_3–Cl, (B) Na + K–Cl and (C) Na + K–HCO_3. Water type A consists of groundwater samples which are a mixed cation facies - Ca^{2+} + Mg^{2+} + Na^+ + K^+ and mixed anion facies mainly HCO_3^- + Cl^-. The cation facie has 40–80% Ca, 10–35% Mg and 30–50% Na + K, whereas, the composition of the anion facie is 60 to nearly 90% HCO_3 and 10–40% Cl. Water type B consists of lake water and has a cation facie of Na + K between 90 and 100% and an anion facie dominated by Cl of 60–70%. Water type C consists of all stream samples with a cation facie of Na + K ranging from 70 to nearly 90% and an anion facie of predominantly HCO_3 (> 80%). Water type A (groundwater) exceeds the WHO (2008) and MBS (2005) MPL for TDS (Fig. 2a). Some of the groundwater samples also exceed the WHO (2008) and MBS (2005) MPL for HCO_3^- (Fig. 3d), F^- (Fig. 3a) and NO_3^- (Fig. 3c). Water type B (lake samples) exceed the WHO (2008) and MBS (2005) MPLs for TDS, Na^+, K^+ F^- and Cl^- (Fig. 2a, b, c; Fig. 3a and b). Water type C (stream samples) do not exceed any of the WHO (2008) and MBS (2005) MPL's. The ionic proportions in groundwater, lake water and stream samples are different because the ions have different sources and/or evolutionally pathways. Because the different water types exist within the Lake Chilwa Basin, it is unlikely that the sources of the ions are related to differences in rock types, rather by secondary processes that affect the chemical evolution of the water.

We developed a conceptual cross section of the groundwater flow in the Lake Chilwa Basin (Fig. 5). There are two main aquifers: a deep fractured basement aquifer and the shallow surficial saprolite/alluvial aquifer. Recharge from rain recharges both the deep and shallow aquifers. Flow is from higher elevations to lower elevations with groundwater discharge from both the shallow and deep aquifers to Lake Chilwa. Streams are supplied from the shallow saprolite groundwater aquifer. Lake Chilwa which is recharged by the deep fractured basement aquifer and the sallow saprolite aquifer is subjected to evaporation. We have shown the water properties that exceed the WHO (2008) and MBS (2005) MPLs in each aquifer in Fig. 5.

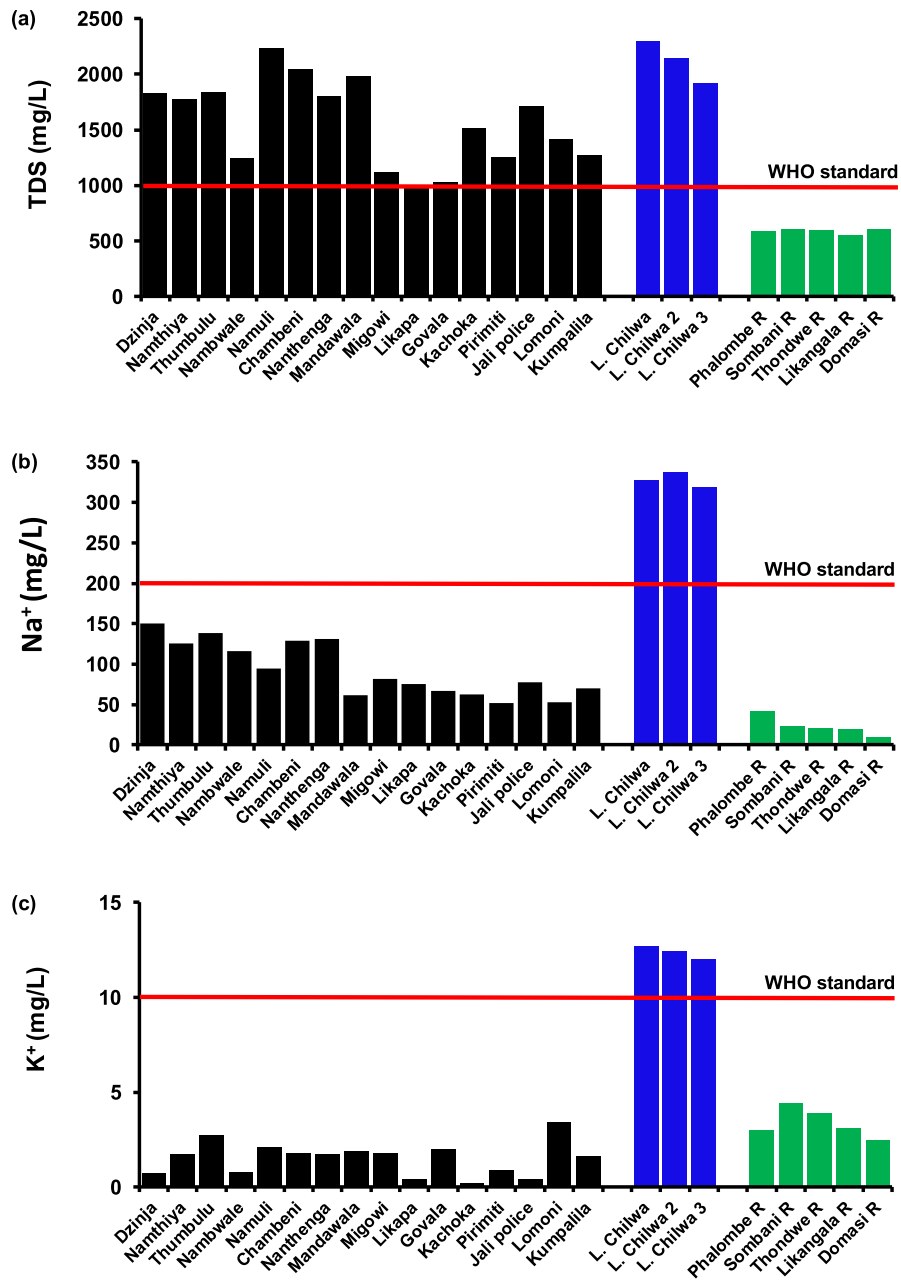

Fig. 2. Plots of concentrations of (a) total dissolved solids (TDS) cations (b) Na^+ and (c) K^+ for groundwater and surface water in the Lake Chilwa Basin. Also shown on the plots is a line representing World Health Organization (WHO) maximum permissible concentrations in drinking water (WHO, 2008).

5.2.2. Cause of high TDS concentrations

The TDS concentrations in natural waters is assumed to be a represent the sum of the cations and anions as reflected in their ability to conduct an electrical current (e.g., Hem, 1985). The major ions in groundwater, lake water and streams can originate from water-rock interaction involving weathering of the silicate rocks in the Lake Chilwa Basin. Because the Lake Chilwa Basin is a closed basin, groundwater in the aquifers at higher elevations flows toward the lake located at the lowest elevation and discharge into the marshes and the lake (Macuiane et al., 2011, Fig. 5). The much higher TDS and Cl^- concentrations in Lake Chilwa compared to groundwater can result from evapo-concentration in the lake (Macuiane et al., 2011). Water-rock interaction and evapo-concentration can be distinguished by evaluating the water for evidence of evaporation. This evidence occurs in the form of enriched stable isotopes of hydrogen and oxygen of water. The δD and $\delta^{18}O$ composition of the groundwater, lake and stream samples are shown in Fig. 6a. We also show the Global Meteoric Water Line (GMWL; Craig, 1961). We constructed a local evaporation water line (LEWL) based on the stream samples and lake samples which are exposed to the atmosphere, and are likely to be affected by evaporation. The LEWL line has the least squares regression defined by the equation: $\delta^2H = 5.7\delta^{18}O + 1$, with a regression coefficient (R^2) of 0.997. The deviation of the lake samples and some stream samples from the GMWL indicates that evaporative enrichment has occurred, resulting in the slope of 5.7 (Fig. 6a) which is less than the GMWL slope of 8. The δD and $\delta^{18}O$ for groundwater samples are more depleted (−39 to −28‰; −6.3 to −4.7‰) than stream samples (−30 to 17; −5.0 to 2.8‰) and lake samples (33 to 45‰; 5.5 to 8.0‰). The more enriched δD and $\delta^{18}O$ water of lake samples and stream samples that lie on the LEWL (Fig. 6a) are affected by evaporation, while groundwater samples show no

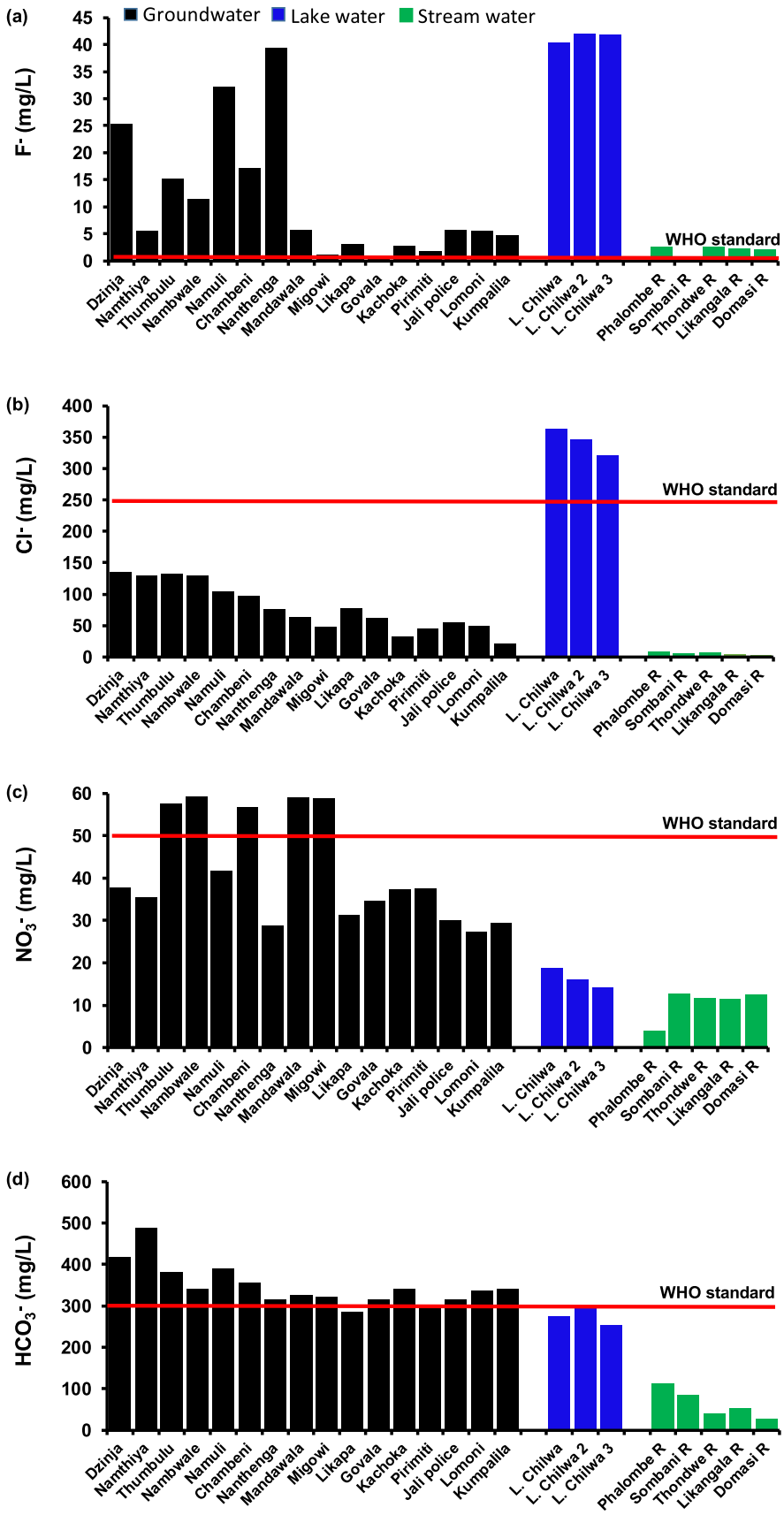

Fig. 3. Plots of concentrations of (a) F^-, (b) Cl^-, (c) NO_3^- and (d) HCO_3^- for groundwater and surface water in the Lake Chilwa Basin. Also shown on the plots are lines representing World Health Organization (WHO) maximum permissible concentrations in drinking water (WHO, 2008).

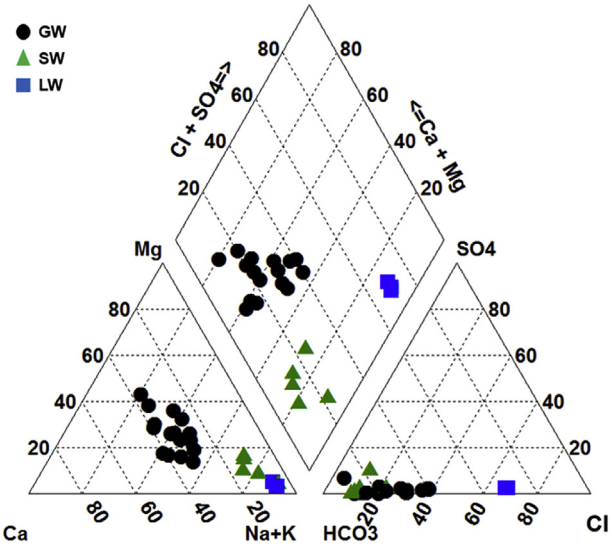

Fig. 4. Piper plot showing ionic proportions for groundwater and surface water and lake water samples in the Lake Chilwa Basin.

evidence of evaporation, as these samples plot on GMWL (Fritz and Clark, 1997).

To evaluate the effect of evaporation on evapo-concentration of solutes, we plot the TDS vs. $\delta^{18}O$ (Fig. 6b). Enriched $\delta^{18}O$ and high TDS can be attributed to evapo-concentration (e.g., Gibilla et al., 2010; Atekwana et al., 2016). There is evidence of evapo-concentration in lake samples from the positive relationship between (TDS) vs. $\delta^{18}O$ where TDS (2000 and 2300 mg/L) and ^{18}O (−5.5 to 8.0‰) are much higher than in stream and groundwater samples. However, no evaporation effect was observed in groundwater samples because TDS concentrations varied from 1000 to 1800 mg/L, while the ^{18}O varied over a small range between −4.7 and −6.6‰. Hence, no evidence of evapo-concentration effect on the groundwater solutes.

Our results indicate that groundwater with high TDS and low $\delta^{18}O$ results from water-rock interaction. Chilton and Smith-Carington (1984) have attributed groundwater with high electrical conductivity (or TDS) to slow movement and more extensive reaction with the crystalline basement aquifer rocks and/or incomplete mixing with fresh groundwater in the shallow alluvial aquifer. Finally, we argue that the low TDS concentrations in the stream samples are because the streams are recharged by recent rains as overland flow and/or freshwater from the shallow weathered regolith/alluvium aquifer. The water in the regolith/alluvium aquifer is fresh because of these aquifers are depleted of highly weatherable minerals and/or rain recharge has a short residence time in the aquifers, thus less extensive water-rock interaction.

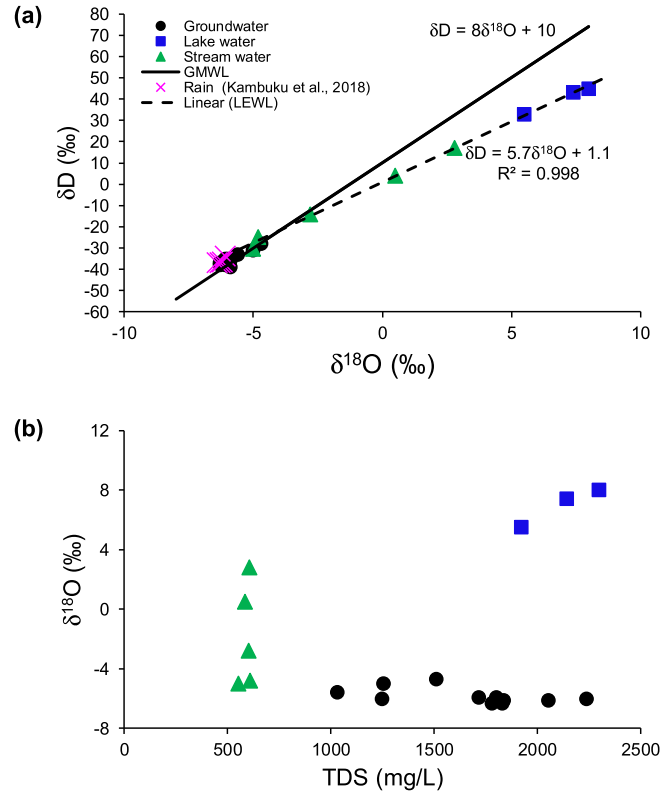

Fig. 6. Cross plot of (a) the stable isotopic composition of oxygen ($\delta^{18}O$) vs. hydrogen (δD). Also plotted is the Global Meteoric Water Line (GMWL) and the local evaporation water line (LEWL), (b) Cross plot of the total dissolved solids (TDS) vs. $\delta^{18}O$ for groundwater and surface water in the Lake Chilwa Basin.

5.2.3. *Cause of high Cl^- concentrations*

The lake samples that exceeded the WHO (2008) and MBS (2005) MPLs are characterized by comparatively high Cl^- concentrations. The presence of Cl^- in groundwater and surface waters is usually attributed to atmospheric sources and trace impurities in rocks and minerals (Freeze and Cherry, 1979) or to anthropogenic pollution. Because only lake samples show elevated concentrations of Cl^-, it is unlikely that the chloride is from rock or mineral weathering or evaporite dissolution. The origin and evolution of Cl^- in natural waters can be evaluated

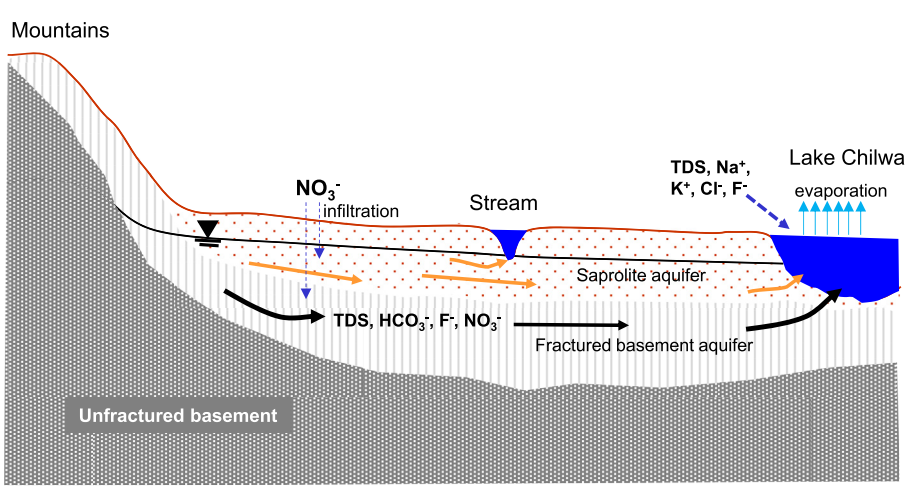

Fig. 5. Cross section of the Lake Chilwa Basin showing the fractured basement aquifer and the surficial saprolite/alluvial aquifer. Also shown are the chemical species that exceed the WHO (2008) and MBS (2005) water quality guidelines. Long arrows in the groundwater aquifers show the direction of groundwater flow. Stream water is supplied from the shallow groundwater aquifer. Lake Chilwa is recharged by the deep fractured basement aquifer and the sallow aquifer and subjected to evaporation.

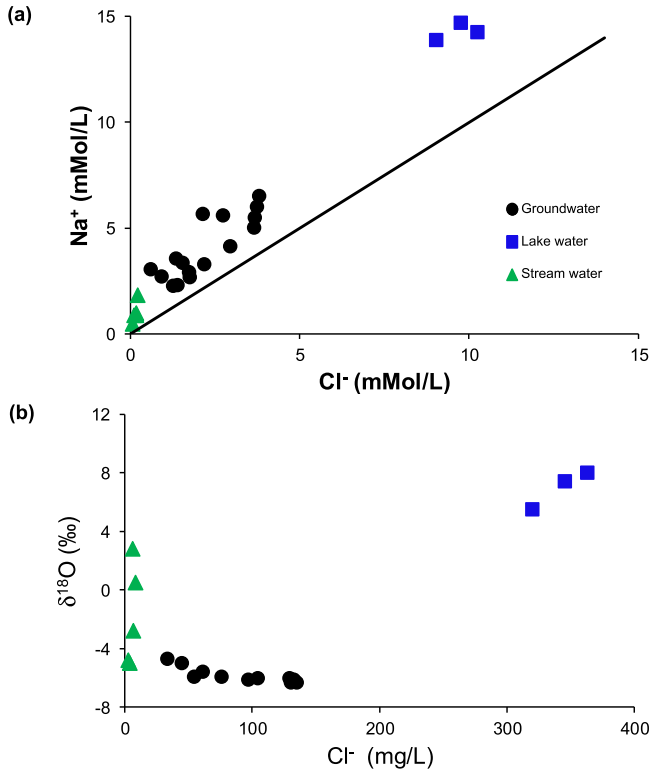

Fig. 7. Cross plot of (a) Cl^- vs. Na^+ and (b) Cl-vs. stable oxygen isotopes ($\delta^{18}O$) for groundwater and surface water in the Lake Chilwa Basin.

using the relationship between Cl^- vs. Na^+ (Fig. 7a). Cl^- is a conservative ion in natural waters and if the origin of Cl^- and Na^+ is from atmospheric fallout or evaporite dissolution, then the ratio of Cl^-:Na^+ is 1:1 (Hem, 1985). The Cl^- vs. Na^+ plot (Fig. 7a) shows that Na^+ concentrations are higher compared to Cl^- concentrations, even though the Cl^- and Na^+ are positively related. We use this positive relationship between Cl^- vs. Na^+ to suggest that the Cl^- in stream, groundwater and lake samples may have the same origin. The origin of Na^+ and Cl^- is from rainfall. The high Na^+ concentrations compared to Cl^- concentrations is consistent with ion exchange buffering in the aquifer (e.g., Neal and Kirchner, 2000).

To evaluate the effect of evaporation on evapo-concentration of Cl^-, we plot the Cl^- concentrations vs. $\delta^{18}O$ (Fig. 6). The $\delta^{18}O$ variations in streams samples are unrelated to Cl^- concentrations, an indication that there are no effects of evaporation on stream samples. In groundwater, the Cl^- concentrations decrease slightly with slight decreases in $\delta^{18}O$ (Fig. 7b), indicating that the Cl^- concentrations in groundwater are not controlled by evaporative enrichment, but rather water-rock interaction. Unlike the case of low Cl^- concentrations in groundwater, the Cl^- concentrations in lake water are high (263–345 mg/L) and positively related to $\delta^{18}O$ (−5.5 to 8.0‰) indicative of evapoconcentration. The concentrations of Na^+, K^+, F^-, Cl^-, SO_4^{2-} are generally higher in the lake water than in groundwater and stream water because Lake Chilwa is a solute sink with no outlet and is subjected to evaporation (Basin and Saka, 2006). Evaporation leads to increased concentration of salts (e.g., NaCl) making the water saltier (Morgan and Kalk, 1970; Basin and Saka, 2006).

5.2.4. Cause of high HCO_3^- concentrations

Only groundwater samples in the Lake Chilwa Basin exceeded the HCO_3^- concentrations stipulated in the WHO (2008) and MBS (2005) MPLs. The HCO_3^- concentrations in natural groundwater is controlled by carbonate evolution (e.g., Drever, 1997). However, it appears that carbonate evolution of groundwater, lake water and stream water in the Lake Chilwa Basin is unlikely to be influenced by the weathering of carbonates. Carbonates are not abundant in the aquifer rocks and minerals, nor are they a major component of the rocks in the basin. Nevertheless, during the weathering and evolution of groundwater in aquifers dominated by silicates, carbonate evolution influences how the water composition changes. For example, feldspar reacts with $CO_{2(g)}$ in recharging meteoric water and forms an insoluble residual Al-silicate mineral and a Ca–HCO_3 water. The weathering of the feldspar proceeds until the solution is saturated with respect to the carbonate minerals (calcite and dolomite). Thus, plagioclase weathering reaction (Eq. (1)) and the calcite equilibrium (Eq. (2)) occur simultaneously:

$$\underset{\text{Anorthite}}{2H_2O + CO_2 + CaAl_2Si_2O_8} \rightarrow \underset{\text{Kaolinite}}{Al_2Si_2O_5(OH)_4} + CaCO_3 \quad (1)$$

where the $CaCO_3$ is formed by the reaction:

$$Ca^{2+} + 2HCO_3^- \rightarrow CaCO_3 \quad (2)$$

The supersaturation of groundwater with respect to carbonates causes calcite to precipitate in fractures and cavities. Secondary calcite is present in fractures, veins and cavities of the granite and gneisses in the Lake Chilwa Basin (Stober and Bucher, 1999).

The saturation state of the groundwater, lake water and stream water with respect to carbonate minerals calcite and dolomite are shown in Fig. 8a and b. As the concentrations of Ca^{2+} or $Ca^{2+} + Mg^{2+}$ increase in groundwater, the groundwater approaches equilibrium saturation with respect to calcite (Fig. 8a) and dolomite (Fig. 8b). Only groundwater samples appear to be close to equilibrium with both calcite (Fig. 8a) and dolomite (Fig. 8b). The carbonate evolution to near saturation with respect to carbonate minerals is responsible for the HCO_3^- concentrations exceeding the WHO (2008) and MBS (2005) MPLs and thus is from natural ionic evolution in groundwater. In fact,

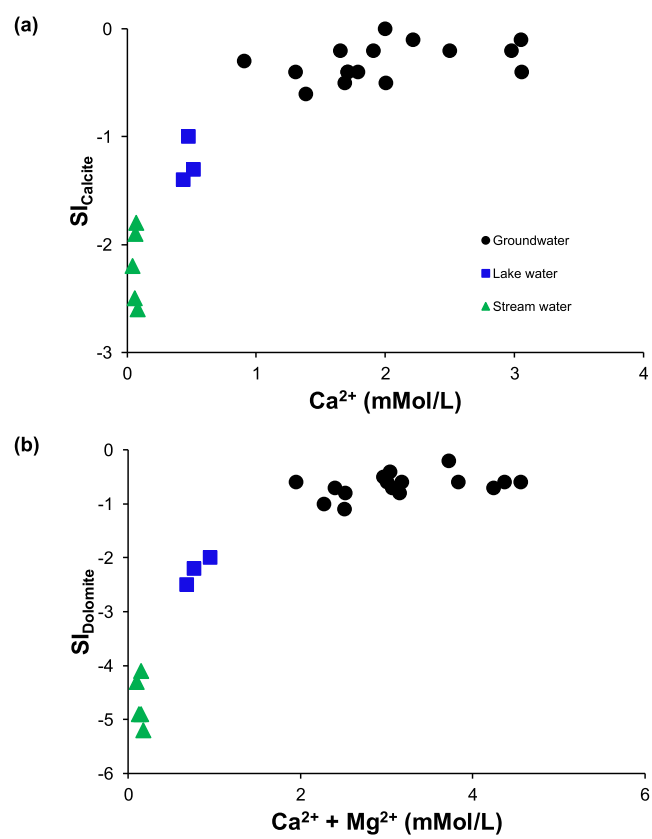

Fig. 8. Cross plots of (a) Ca^{2+} vs the saturation index for calcite ($SI_{calcite}$) and (b) $Ca^{2+} + Mg^{2+}$ vs. saturation index for dolomite ($SI_{dolomite}$) surface water and lake water samples in the Lake Chilwa Basin.

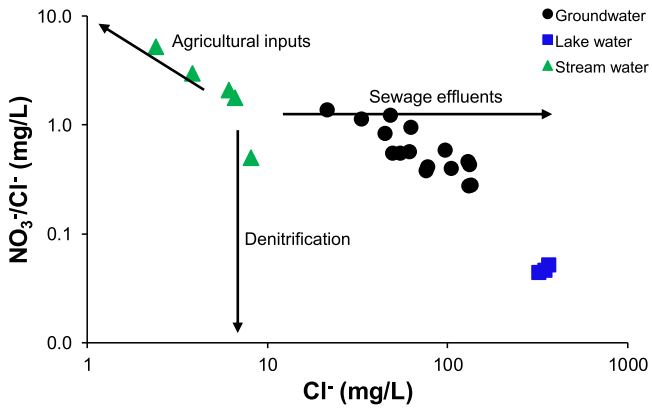

Fig. 9. Cross plot of Cl^- vs. NO_3^-/Cl^- for groundwater, stream and lake samples in the Lake Chilwa Basin.

stream water is relatively undersaturated with respect to carbonates compared to groundwater in the fractured basement aquifer. We interpreted the saturation state of stream water to mean that the groundwater in the regolith/alluvial aquifer which supplies water to streams chemically evolves independent of groundwater in the deep fractured basement aquifer.

5.2.5. Cause of high F^- concentrations

The elevated concentrations of F^- in groundwater in the Lake Chilwa Basin that exceed the WHO (2008) and MBS (2005) MPLs may be attributed to the dissolution of the mineral apatite (Carter and Bennet, 1973). The concentrations of F^- are some of the highest that have been observed in groundwater in Malawi (BGS, 2004; Msonda et al., 2007, Sajidu et al., 2007). The high F^- concentrations in stream and groundwater samples can be attributed to weathering of apatite, biotite, hornblende and amphibole in granitic and gneissic rocks in the Lake Chilwa Basin (e.g., Carter and Bennet, 1973; Msonda et al., 2007, Sajidu et al., 2007). The F^- concentrations in groundwater are much higher compared to river water because of the longer residence of groundwater and greater water-rock interaction in the fracture crystalline basement aquifer. The high concentrations of F^- in Lake Chilwa is attributed direct input of F^- from streams and groundwater and from evapoconcentration in the lake (Fig. 5). We speculate that the low concentrations of F^- in streams may result from stream recharge coming from the shallow aquifer composed of weathered regolith and alluvium depleted in apatite, biotite, hornblende and amphibole which are the sources of F^- in the Lake Chilwa Basin.

5.2.6. Cause of high NO_3^- concentrations

The high concentrations of NO_3^- that exceed the WHO (2008) and MBS (2005) MPLs may be due to decaying organic matter or improper disposal of sewage and contamination from excess fertilizers used in the catchment (Karnath and Hartje, 1987). A plot of NO_3^- vs. Cl^- (Fig. 9) is useful to distinguish between the various sources of nitrate in the basin (e.g., Anornu et al., 2017). It appears that the source of NO_3^- in groundwater and lake water is from sewage effluent from pit latrines. The NO_3^- in surface water although not above the WHO (2008) and MBS (2005) MPL is from fertilizers application associated with agriculture.

6. Conclusions

We conducted a geochemical investigation of groundwater, stream water and lake water in the Lake Chilwa Basin in southern Malawi. Our goal was to assess water quality and determine the processes that controlled water quality. All groundwater samples exceed the WHO (2008) and MBS (2005) MPLs for TDS and some of the groundwater samples exceed the WHO (2008) and MBS (2005) MPL for HCO_3^-, NO_3^- and F^-. All lake samples exceed the WHO (2008) and MBS (2005) MPLs for TDS, Na^+, K^+, Cl^- and F^-. No stream samples exceed any of the WHO (2008) and MBS (2005) MPL's. Hydrochemical analyses revealed that groundwater quality in the fracture basement aquifer was controlled by water-rock interaction except for NO_3^- which was from anthropogenic pollution. Water from Lake Chilwa derived from streams and groundwater was evaporated, which enriched solutes; evapoconcentration caused or enhanced poor water quality. The quality of stream water was good. Stream water sources include overland flow during the rainy season and shallow groundwater discharge during baseflow. Water-rock interactions within the shallow regolith/alluvial aquifer that support streams does not adversely affect water quality. We surmised that poor groundwater quality for drinking and domestic use is due to natural water-rock interactions and anthropogenic pollution. Because stream water is of good quality, and because baseflow to streams is supported by groundwater from the regolith/alluvial aquifers, efforts should be made to map fresh groundwater aquifers for domestic, agricultural and industrial use.

Acknowledgements

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